KEVIN WILLIAMS

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TECHNICAL SKILLS

Mechanical Engineering: Metallography, Microscopy, Materials Characterization, Tensile/Hardness Testing, Failure Analysis, CAD, GD&T, Test Fixture Design, DFMEA/PFMEA

Systems Engineering: Requirements Development & Verification (NPR 7123, NPR 7150.2), MBSE, V&V, Systems Lifecycle, Data Architecture

AI/ML: PyTorch, TensorFlow, LLMs, RAG, Transformers, Hugging Face, CLIP, FAISS, LangChain, AI Certification & Governance, Autonomous System Verification

Software: Python, SQL, JavaScript, Django, Flutter, MySQL, Git, AWS, FastAPI, Docker

Methods: Black Belt Six Sigma, Differential Privacy, Adversarial Robustness, Red-Teaming

PROFESSIONAL COMPETENCIES

Human Spaceflight Systems Engineering • Materials & Test Engineering • Commercial and Government Spaceflight Integration • Cross-Functional Technical Leadership • Multi-Program Integration

EXPERIENCE

Systems Engineer — Barrios Technology (NASA Johnson Space Center) Oct 2023 – Feb 2025, Jan 2026 – Present

- Developed, reviewed, and verified systems engineering requirements across ISS, Orion, and Commercial Lunar Delivery Program (CLDP), including interface and safety requirements for commercial partner integration, detailed and final test objectives (DFTOs/FTOs), and flight test data evaluation
- Applied systems engineering lifecycle processes (NPR 7123) to human spaceflight systems, leveraging mechanical engineering background to assess requirements, verification approaches, and failure modes across subsystems including ECLSS, propulsion, and structural systems
- Integrated AI requirements framework into CLDP, establishing evaluation criteria and verifiable compliance pathways for autonomous systems operating in safety-critical human spaceflight environments
- Defining data architecture for the Extravehicular Activity and Human Surface Mobility Program (EHP), establishing data standards, schemas, and tooling infrastructure to consolidate program data management across teams
- Coordinated with commercial partners including Axiom Space to align requirements, resolve interface challenges, and ensure compliance with NASA human spaceflight safety standards

Software Engineer — Barrios Technology Feb 2025 – Dec 2025

- Architected and deployed Cortex, a unified enterprise data platform with integrated LLM capabilities for Intuitive Machines, replacing fragmented workflows with a single interface for document management, data retrieval, and cross-team collaboration
- Developed cross-platform Flutter application for KULR Technologies automating report generation and operational workflows

Engineering Specialist — Ford Motor Company Jun 2022 – Sep 2023

- Led 20-person eMotor materials laboratory, performing and directing metallurgical cross-sectioning, microscopy, hardness, and tensile testing to characterize material properties and qualify components for electrified powertrain production programs
- Conducted failure analysis and root cause investigations on eMotor components, contributing to DFMEA/PFMEA processes and driving design iterations to resolve material degradation, delamination, and manufacturing defects
- Designed and specified test fixtures, tooling, and measurement systems using CAD with GD&T to support materials characterization and validation testing across eMotor programs
- Developed data collection and automated inspection systems to capture production quality metrics across the eMotor lifecycle, feeding quantitative measurement data back into design reviews to drive component and process improvements
- Coordinated with tier suppliers and cross-functional teams across design, manufacturing, and quality engineering to review part drawings, qualify materials, execute supplier corrective actions, and resolve hardware integration challenges across electrified vehicle programs

Research Assistant — University of Michigan Sep 2019 – Jun 2022

- Developed novel electrochemical detection method for Δ9-THC (1–20 µM) achieving 0.13 µM limit of detection with $R^2 = 0.995$, third-lowest detection limit among comparable SPCE devices
- Published peer-reviewed findings demonstrating viability for in-field law enforcement applications

Materials & Chemical Technician — Hyundai-Kia, General Motors, Quaker Chemicals 2012 – 2017

Metallurgical analysis, material validation, and quality testing across automotive OEMs. Established GM Flint Engine metallurgical lab; led supplier corrective actions at Hyundai-Kia; trained personnel on QA protocols.

RESEARCH PROJECTS

Privacy-Preserving Cognitive Twin: Membership Inference Attacks & Defenses M.S. AI — CIS 545

- Designed complete adversarial evaluation pipeline studying model vulnerability to membership inference attacks (confidence-based, label-only, and shadow model variants)
- Implemented and benchmarked three defense strategies, DP-SGD (via Opacus), Model Confidence Exclusion, and a fusion defense, sweeping ε values to empirically map privacy–utility tradeoffs
- Produced structured evaluation coverage analysis comparing baseline vs. defended models across attack success rate, AUROC, accuracy, and calibration metrics

µStruct — AI-Powered Microstructure Identification System ASTM Project

- Built ensemble classifier combining CLIP zero-shot inference with FAISS similarity-weighted voting across 6,000+ micrographs, live demo: kevwill-microstructure-classifier.hf.space
- Designed and validated grading accuracy methodology evaluating ensemble weighting (40/60 zero-shot/similarity split) against ground-truth phase labels across 16 steel microstructural phases
- Productionized into FastAPI backend with Docker deployment on Hugging Face Spaces; REST API for retrieval, classification, and evaluation

Disaster Infrastructure Damage Recognition via Computer Vision M.S. AI — CIS 583

- Developed CNN-based binary classifier (disaster/non-disaster) for infrastructure damage assessment in post-event aerial and ground imagery
- Conducted empirical evaluation of model performance metrics to establish classification confidence thresholds for first-responder deployment readiness

EDUCATION

M.S. in Artificial Intelligence — University of Michigan (May 2026)
B.S. in Mechanical Engineering — University of Michigan (December 2021)

PUBLICATIONS

Williams, K. *Requirements and Verification Standards for Artificial Intelligence in Safety-Critical Applications.* Safety Critical Labs. Zenodo, 2026. DOI: 10.5281/zenodo.19323435
Differential pulsed voltammetry of Δ9-THC on disposable screen-printed carbon electrodes: A potential in-field method to detect Δ9-THC in saliva (2023)

AWARDS

Ford Technical Excellence: Implementation of Deep Learning Model Technology for Quality Decisions
Silver Bear: Certification of Artificial Intelligence on NASA Human-Rated Space Flight Systems